MIL-S-3786/17B
17 January 1972
SUPERSEDING
MIL-S-3786/17A
5 November 1965

MILITARY SPECIFICATION

SWITCH, ROTARY, ELECTRO-MECHANICAL

ACTUATED (SOLENOID TYPE),

STYLE SR17

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

• DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

1. SCOPE

1.1 This specification covers the detail requirements for style SR17 electro-mechanical actuated rotary switches.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-3786 - Switches, Rotary (Circuit Selector, Low-Current Capacity), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

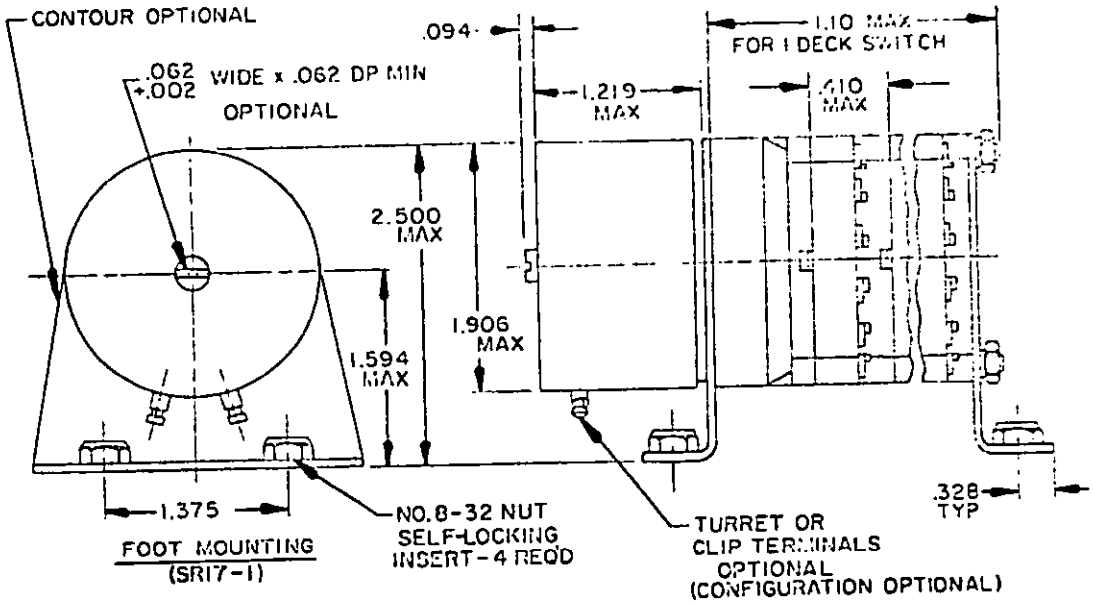
3.1 Requirements. Requirements shall be in accordance with MIL-S-3786, and as specified herein.

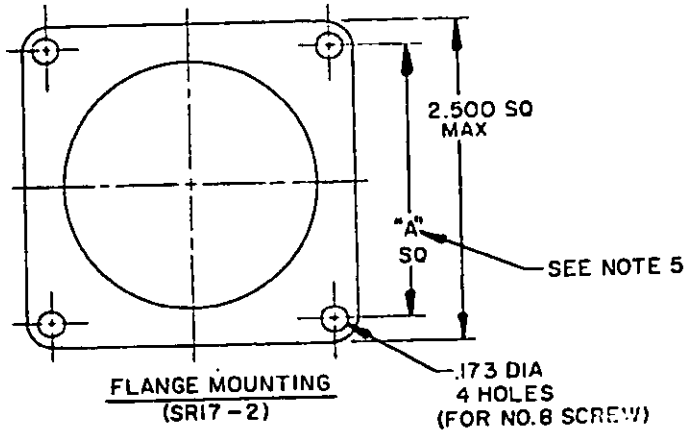
3.2 Design and construction. Switches shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Switch terminals. Terminals shall project in a direction parallel to the plane of the section.

3.3 Solenoid-input voltage. Solenoid-input voltage shall be 24 Vdc minimum, 30 Vdc maximum.

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INCHES	MM
.002	.05
.060	1.52
.062	1.57
.094	2.39
.173	4.39
.328	8.33
.410	10.41
1.075	27.31
1.10	27.94
1.219	30.96
1.375	34.93
1.594	40.49
1.906	48.41
2.500	63.50

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances shall be ±1/64(.40 mm) on fractions and ±.005 (.13 mm) on decimal dimensions.
4. Interrupter switch mechanism may extend below dotted line, but not below mounting feet.
5. 'A' dimensions shall be 1.250(31.75 mm) or 2.125(28.58 mm) as specified.

FIGURE 1. Style SR17 solenoid operated switch.

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3.4 Solenoid-input current. Maximum-input current shall not exceed 5 amperes at 85°C or 3 amperes at 125°C.

3.5 Terminal strength (pull). A force of 3 pounds shall be applied to the switch terminals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-3786, and as specified herein.

4.2 Life (rotational). Test loads and applicable-circuit conditions shall be as specified in table I. Each of the loads specified for the applicable-environmental condition shall be switched by at least one rotor contact of the switch.

TABLE I. Circuit values for life (rotational) test.

Environmental condition	Inductive load (2.8 henries)		Resistive loads	
	Milli-amperes	Volts, dc	Milli-amperes	Volts
At atmospheric pressure - - - - -	100	30	200 50	30 dc 150 ac
At reduced barometric pressure -	25	30	200 20	30 dc 150 ac

4.2.1 Stepping speed. Solenoid-coil current shall be interrupted at a rate sufficient to cause switch deck contacts to be stepped at the rate of 20 steps per second, minimum.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-3786.

6. NOTES

6.1 The notes specified in MIL-S-3786 are applicable to this specification.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Number of switch positions, angle of throw, and electrical configuration.
(b) Homing (control) deck and configuration, as required.
(c) Flange or foot mounting (see figure 1).
(d) Solenoid input voltage range and power source type.
(e) Solenoid duty cycle. Specify maximum "ON" and minimum "OFF" times.
(f) Flange mounting dimension, if applicable (see figure 1).
(g) Lead wires and length (when required).

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NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI
Navy - EC, AS, OS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, AT, AV, SM
Navy - MC, CG
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5930-0831)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of ification in procurement of products for ultimate use by the Department of Defense. This sheet is provide taining information on the use of this specification which will insure that suitable products can be procur inimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there ar tional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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